

Short Questions and Answers (According to Slides)

Greedy Technique

Q1. What is a greedy algorithm?

A greedy algorithm is a problem-solving technique that chooses the best available option at each step without considering future consequences.

Q2. In which type of problems are greedy algorithms mainly used?

Greedy algorithms are mainly used in optimization problems.

Q3. What is the Greedy Choice Property?

It means the optimal solution can be achieved by making the locally best choice at every step.

Q4. What is Optimal Substructure?

Optimal substructure means the optimal solution contains optimal solutions of its subproblems.

Q5. Why do greedy algorithms fail in some cases?

Because choosing the locally optimal solution may not always produce the globally optimal solution.

Q6. Give examples where greedy algorithms fail.

- Coin Change Problem
- 0/1 Knapsack Problem

Q7. What are the applications of greedy algorithms?

- Fractional Knapsack
 - Job Scheduling
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Fractional Knapsack Problem

Q8. What is the Fractional Knapsack Problem?

It is a problem where items can be taken fractionally to maximize profit without exceeding capacity.

Q9. Which greedy strategy gives the best result in Fractional Knapsack?

Selecting items based on the profit/weight ratio gives the best result.

Q10. Can items be divided in Fractional Knapsack?

Yes, items can be taken partially or fractionally.

Minimum Spanning Tree (MST)

Q11. What is a spanning tree?

A spanning tree is a subgraph that contains all vertices of a graph and forms a tree.

Q12. What is a Minimum Spanning Tree?

A Minimum Spanning Tree is a spanning tree with the minimum total edge cost.

Q13. Name two MST algorithms.

- Kruskal's Algorithm
- Prim's Algorithm

Q14. How many edges does an MST contain?

An MST contains $n-1$ edges where n is the number of vertices.

Kruskal's Algorithm

Q15. What is Kruskal's Algorithm?

Kruskal's Algorithm is a greedy algorithm used to find the Minimum Spanning Tree by selecting minimum cost edges without forming cycles.

Q16. What data structure is commonly used in Kruskal's Algorithm?

Priority Queue is commonly used.

Q17. What happens if an edge forms a cycle in Kruskal's Algorithm?

The edge is rejected.

Q18. Why is Kruskal's Algorithm called greedy?

Because it always selects the minimum cost edge available.

Single Source Shortest Path (SSSP)

Q19. What is the shortest path problem?

It is the problem of finding the minimum cost path between nodes in a graph.

Q20. What is the cost of a path in a weighted graph?

The cost is the sum of all edge weights along the path.

Q21. Name the variations of shortest path problems.

- Single-pair shortest path
 - Single-source shortest path
 - Single-destination shortest path
 - All-pair shortest path
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Dijkstra's Algorithm

Q22. What is Dijkstra's Algorithm?

It is a greedy algorithm used to find the shortest path from a source node to all other nodes.

Q23. Does Dijkstra's Algorithm work with negative edges?

No, it does not work with negative edge weights.

Q24. Which approach does Dijkstra's Algorithm use?

Greedy approach.

Q25. What is relaxation in Dijkstra's Algorithm?

Relaxation updates the shortest distance if a shorter path is found.

$$d(u) + c(u, v) < d(v)$$

Then:

$$d(v) = d(u) + c(u, v)$$

Q26. What is the initial distance of the source node in Dijkstra's Algorithm?

The initial distance of the source node is 0.

Q27. What is the initial distance of other nodes in Dijkstra's Algorithm?

Infinity (INF).

Bellman-Ford Algorithm

Q28. What is Bellman-Ford Algorithm?

Bellman-Ford Algorithm finds shortest paths in graphs with negative edge weights.

Q29. What is the main operation in Bellman-Ford Algorithm?

Relaxation of edges.

Q30. How many times are edges relaxed in Bellman-Ford Algorithm?

$V-1$ times, where V is the number of vertices.

Q31. How can a negative weight cycle be detected?

If any distance changes in the V -th iteration, then a negative weight cycle exists.

Q32. Does Bellman-Ford work with negative edges?

Yes.

Topological Sort

Q33. What is Topological Sorting?

It is a linear ordering of vertices in a DAG where u appears before v if there is an edge $u \rightarrow v$.

Q34. On which graph is topological sorting possible?

Directed Acyclic Graph (DAG).

Q35. Why is topological sorting impossible in cyclic graphs?

Because dependency ordering cannot be satisfied in cycles.

Q36. Give applications of topological sorting.

- Task scheduling
 - Course prerequisite ordering
 - Build systems
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Kahn's Algorithm

Q37. What is Kahn's Algorithm?

Kahn's Algorithm is a BFS-based algorithm for topological sorting.

Q38. Which vertices are inserted into the queue first in Kahn's Algorithm?

Vertices with indegree 0.

Q39. What is indegree?

Indegree is the number of incoming edges of a vertex.

DFS Based Topological Sort

Q40. What data structure is used in DFS-based topological sorting?

Stack.

Q41. When is a node pushed into the stack in DFS topological sort?

After visiting all its adjacent vertices.

Backtracking

Q42. What is Backtracking?

Backtracking is a technique where solutions are built incrementally and invalid solutions are abandoned.

Q43. What happens when backtracking reaches a dead end?

The algorithm returns to the previous state and tries another path.

Q44. Name some applications of backtracking.

- Sudoku
- N Queens
- Graph Coloring
- Puzzle solving

Q45. What are the main steps of backtracking?

- Choose
 - Explore
 - Check validity
 - Backtrack
 - Repeat
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Graph Coloring Problem

Q46. What is graph coloring?

Assigning colors to graph vertices so that adjacent vertices do not share the same color.

Q47. What is m-coloring?

Coloring a graph using at most m colors.

Q48. What is Chromatic Number?

The minimum number of colors needed to color a graph.

Q49. Give applications of graph coloring.

- Map coloring
 - Timetable scheduling
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N Queens Problem

Q50. What is the N Queens Problem?

It is the problem of placing N queens on an $N \times N$ chessboard so that no two queens attack each other.

Q51. Which positions must queens avoid?

- Same row
- Same column

- Same diagonal

Q52. Which technique is used to solve the N Queens Problem?

Backtracking.

Branch and Bound

Q53. What is Branch and Bound?

It is an optimization technique used to find optimal solutions by branching and pruning.

Q54. What is branching?

Dividing a problem into smaller subproblems.

Q55. What is pruning?

Eliminating branches that cannot produce better solutions.

Q56. What is bounding?

Computing upper or lower limits for subproblems.

Q57. Which search strategy is used in FIFO Branch and Bound?

Breadth First Search (BFS).

Approximation Algorithm

Q58. What is an approximation algorithm?

An algorithm that finds near-optimal solutions for NP-hard problems.

Q59. Why are approximation algorithms needed?

Because exact polynomial-time solutions are not available for NP-hard problems.

Q60. What is an approximate solution?

A feasible solution close to the optimal solution.

Traveling Salesman Problem (TSP)

Q61. What is the Traveling Salesman Problem?

The problem of visiting all cities exactly once and returning to the starting city with minimum cost.

Q62. What type of problem is TSP?

NP-hard problem.

Q63. What is Triangle Inequality?

$$d(A,C) \leq d(A,B) + d(B,C)$$

The direct path should not cost more than an indirect path.

Q64. Which tree is used in TSP approximation?

Minimum Spanning Tree (MST).

Vertex Cover Problem

Q65. What is a vertex cover?

A subset of vertices such that every edge has at least one endpoint in the subset.

Q66. What is the goal of the Vertex Cover Problem?

To find the minimum size vertex cover.

Q67. Which graph is used in Vertex Cover Problem?

Undirected graph.

Q68. Give a real-life application of Vertex Cover.

Traffic camera placement.

